Theorem 1: $a^2 \equiv 1 \pmod{a+1}$ for all $a \in \mathbb{Z}$.

Proof: Since $a^2 - 1 = (a+1)(a-1)$, we have $a^2 - 1 \equiv 0 \pmod{a+1}$. Therefore, $a^2 \equiv 1 \pmod{a+1}$. ■

Corollary (Rocco's Theorem): For prime $p > 3$, the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times \neq \langle p-1 \rangle$.

Proof: By Theorem 1, the order of the element $p - 1 \in (\mathbb{Z}/p\mathbb{Z})^\times$ is 2. The order of $(\mathbb{Z}/p\mathbb{Z})^\times$, however, is $\phi(p) = p - 1 > 2$. It is, therefore, impossible to generate $(\mathbb{Z}/p\mathbb{Z})^\times$ with $p - 1$.